

Smart Wheel Misalignment and Tire Pressure Detection system for Automobiles

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INTRODUCTION

- Wheel alignment and tire pressure are crucial for vehicle safety and optimal performance.
- Traditional methods of alignment and tire checks are expensive and require bulky stationary equipment.
- The wheels has three angles camber, caster and toe which are needed to be within the required range for proper alignment.
- Maintaining the correct tire pressure is essential for stable handling, preventing uneven wear, and ensuring safe vehicle performance.

MOTIVATION

- ❖ Misaligned wheels and improper tire conditions lead to safety concerns, higher maintenance costs, and increased wear and tear.
- ❖ The misaligned wheels causes accidents
- ❖ Traditional alignment checks are expensive and require specialized equipment.
- ❖ Incorrect tire pressure (too high or too low) reduces fuel efficiency, damages the tire structure, and increases the chances of tire bursts.

LITERATURE SURVEY

Sl. No.	Title, Author, Year	Summary	Limitation /Gaps
1.	K.Yaminii,K. Thulasi Priya,D. Archana,Ch. Monika,Sumalatha Emmela, "Detection and Prevention of Wheel Unbalancing and Tire Burst in Moving Vehicles ",2020.	The paper proposes using an Arduino-based system with ultrasonic and DHT-11 sensors to detect wheel misalignment and tire temperature, aiming to prevent accidents caused by bearing failure and tire bursting.	The system relies on traditional methods, which may not be as accurate or timely as automated solutions.
2.	Mohamad Hadi,Sulaiman,Suhana Sulaiman,Azilah Saparon, "IoT for wheel alignment monitoring system",2021.	This paper presents a portable wheel alignment monitoring system using IoT, where sensors and a microcontroller communicate with a mobile app via TCP/IP to provide real-time detection and display of wheel misalignment	The system requires a smartphone and reliable wireless communication, which may limit accessibility in certain environments.
3.	Sharun Mohan P, Abhinav Das,"Automatic wheel Alignment Detector with Alarm System",2023.	This review paper analyzes automatic wheel alignment detectors, comparing optical, laser, and sensor-based technologies, and explores the integration of AI and machine learning to improve accuracy and reduce manual labor.	The integration of AI and machine learning in wheel alignment detection still faces challenges in cost and compatibility.

Sl. No.	Title, Author, Year	Summary	Limitation /Gaps
4.	Aaron Ameerli, Nadine Sangster and Gerard Ragbir, ”Autonomous detection of vehicular wheel alignment parameters “,2020.	This paper presents a framework for a sensing system that alerts drivers when their vehicle’s alignment is off, contributing to the growth of smart vehicle technologies that enhance safety and decision-making.	The proposed alignment detection framework is limited by the availability of 3D vision technology and its complexity.
5	Florian Grimma,Daniel Kiefera, Tim Strauba, Gunter Bitsch , Clemens van Dintherb ,“Automatic Gear Tooth Alignment in Vision Based Preventive Maintenance”,2024.	This paper investigates visual-based preventive maintenance for hobbing wheels, exploring different alignment methods to capture precise images for assessing tool condition, with deep neural networks outperforming traditional algorithms.	Traditional algorithms may not perform as well as deep neural networks, and the system requires significant computational resources.
6	Azuwien Aida Bohari, Muhamad Faris Haikal, Mohammad Hafiz, Sim Sy Yi, Norezmi Jamal, Mohd Nizam ,Mohd Talib, Syadia Nabilah Mohd Safuan,“Development of Automobile Wheel Smart Alignment Monitoring System”,2024.	While the system enhances safety, it may not be effective in all driving conditions, especially in extreme environments	While the system enhances safety, it may not be effective in all driving conditions, especially in extreme environments.

4. Outcome of Literature survey

➤ Our literature review highlights a significant landscape of research in:

- Wheel Misalignment Detection: Explored various vision-based, sensor-based, and AI/ML integrated methods, demonstrating a strong foundation for detection.
- Temperature & Pressure Monitoring: While individual monitoring exists, their integration with misalignment detection for real-time alerting is less common.
- Existing solutions often present challenges in real-time applicability, complexity, or robustness across diverse conditions.
- A crucial gap identified is the lack of a unified, integrated system for simultaneous real-time wheel misalignment detection, temperature, and pressure monitoring, coupled with an immediate alerting mechanism.
- Our project aims to address this by developing a comprehensive, integrated, and real-time solution to enhance vehicle safety and predictive maintenance.

➤ **Key Gaps & Project Focus:**

- Existing solutions often present challenges in real-time applicability, complexity, or robustness across diverse conditions.
- A crucial gap identified is the lack of a unified, integrated system for simultaneous real-time wheel misalignment detection, temperature, and pressure monitoring, coupled with an immediate alerting mechanism.
- Our project aims to address this by developing a comprehensive, integrated, and real-time solution to enhance vehicle safety and predictive maintenance.

PROBLEM STATEMENT

Wheel misalignment affects vehicle safety, handling, and tire wear & tear.

Traditional systems are costly and need professionals.

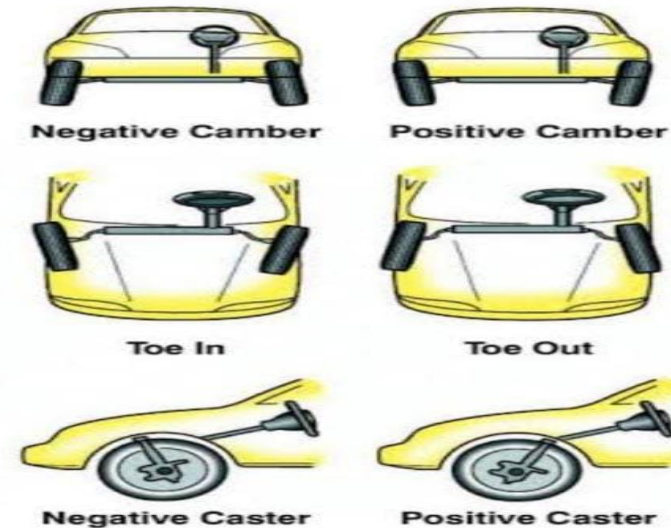
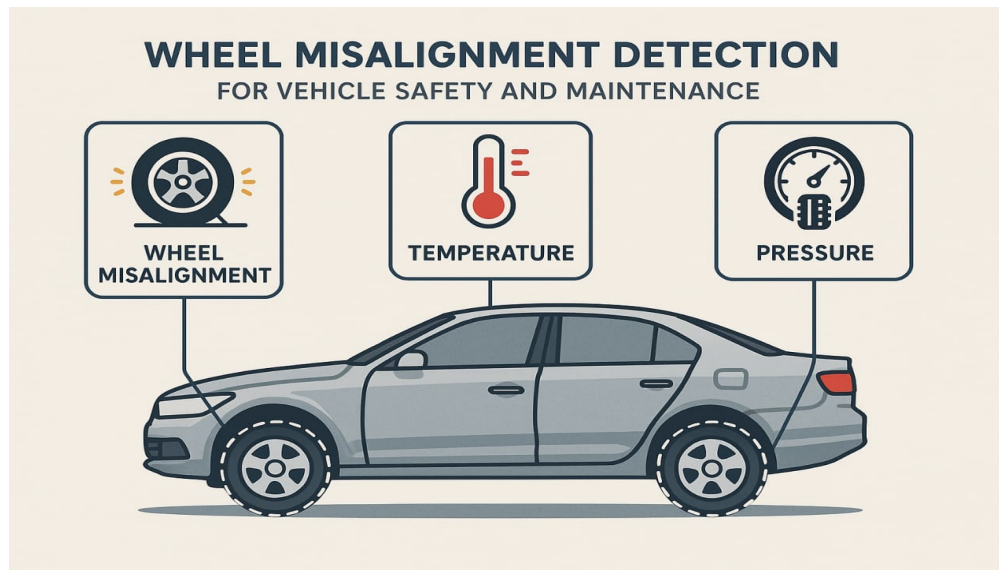


Fig 1 : Alignment angles views

OBJECTIVES

Objective 1: To check alignment for camber, toe, and caster of the wheel

Objective 2: To measure tire pressure.

Objective 3: To provide early warnings for misalignment and unsafe tire conditions, ensuring optimal vehicle performance.

Objective 4: To create a guided tire misalignment detection systems

METHODOLOGY

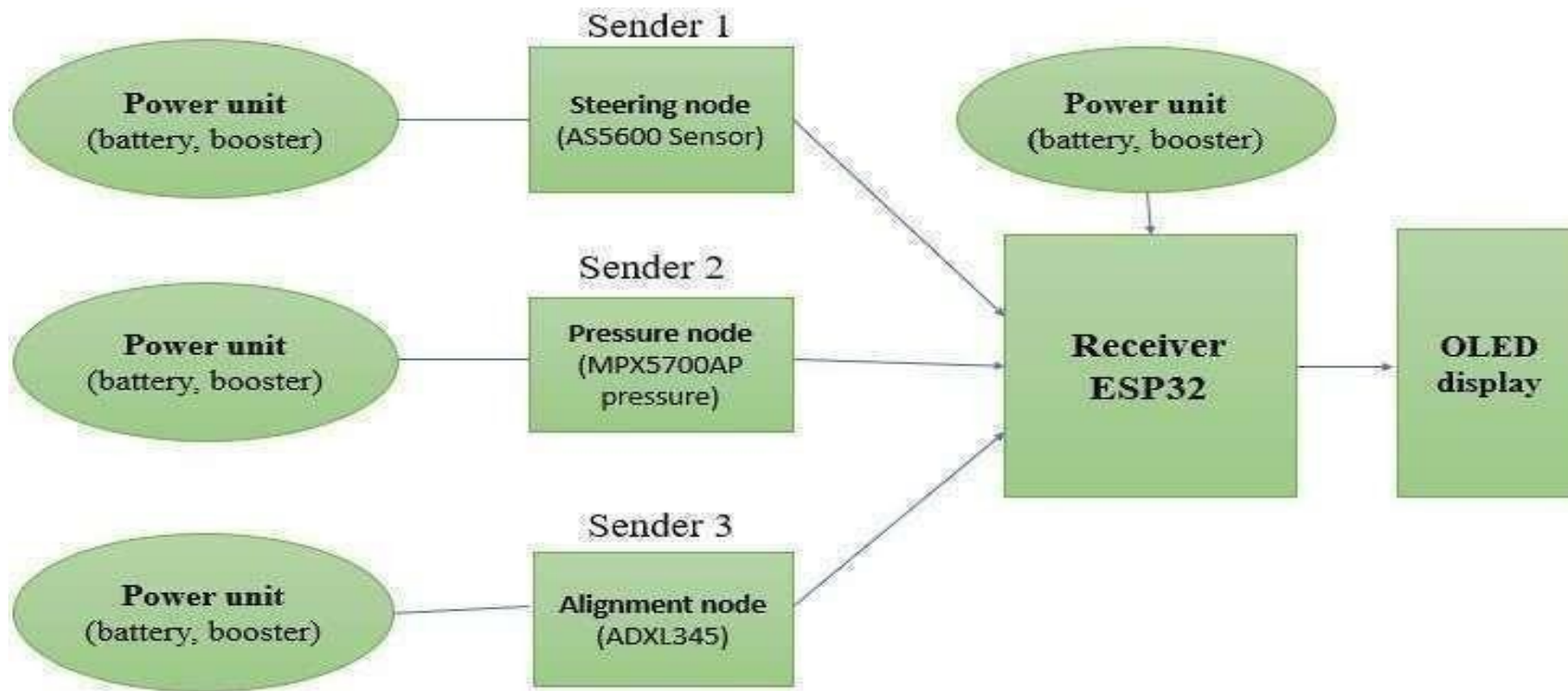
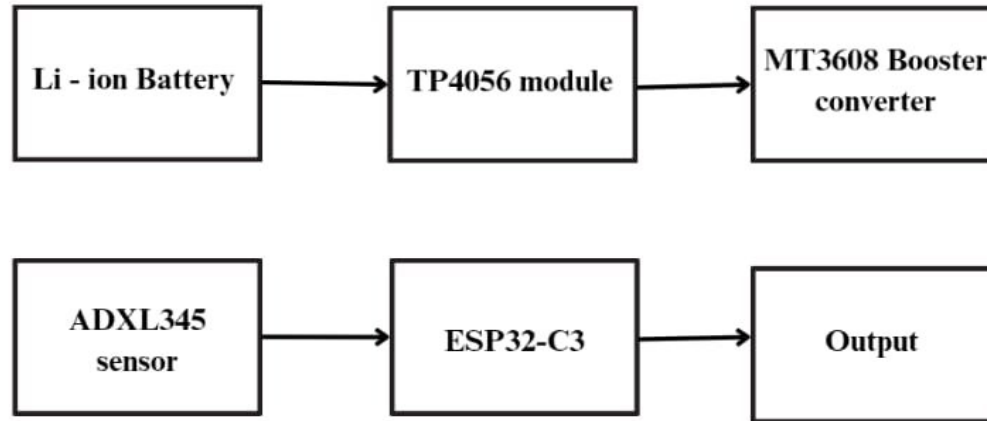
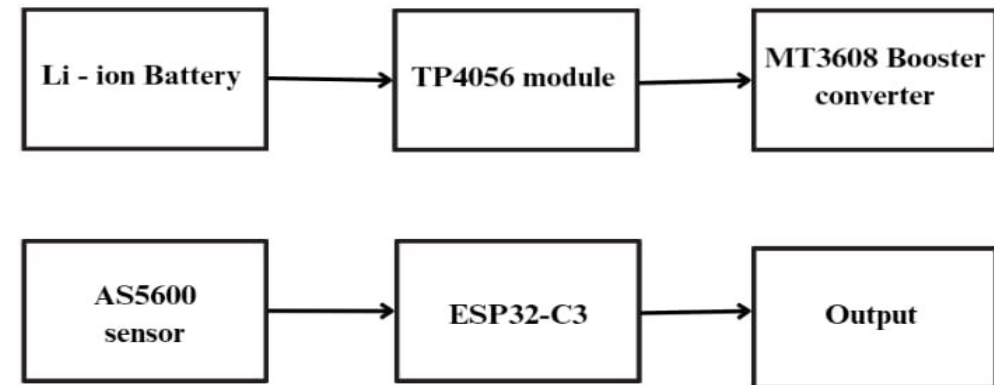


Fig 1 : Block Diagram

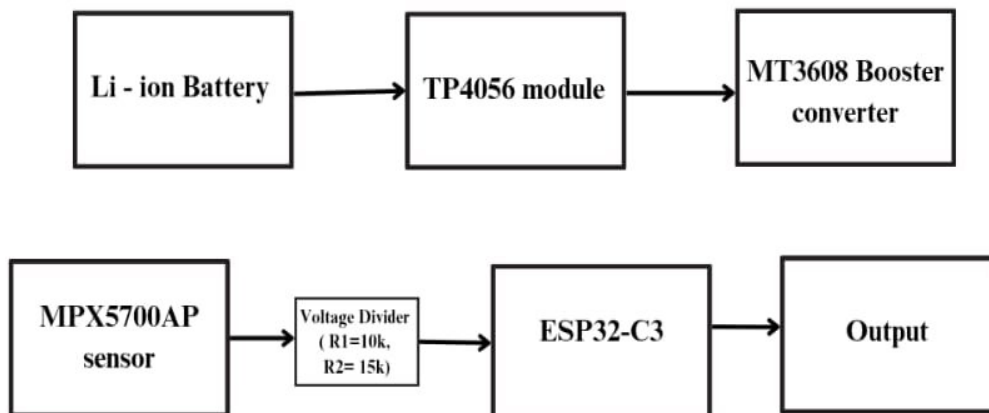
ADXL345 Sender Module



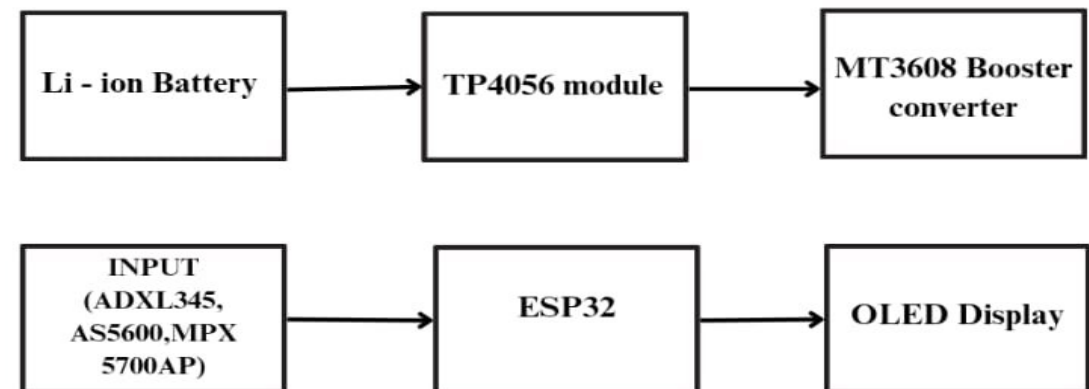
AS5600 Sender Module



MPX5700AP Sender Module



ESP32 Receiver Module



METHODOLOGY

1. Sensor Data Acquisition

- The ESP32C3 collects camber, caster values from ADXL345, steering angle from AS5600, and tire pressure from the pressure sensor. All readings are received at the ESP32 receiver through ESP-NOW wireless communication.

2. Calibration

- The system sets the zero reference for steering (straight-wheel position). A baseline orientation is also set for accurate camber & caster calculation

3. Data Processing

- Each parameter is compared with standard thresholds Camber, Caster, Toe → Alignment standards, Tire Pressure → Safe pressure limits. Real-time calculations determine whether each value is within the allowed range.

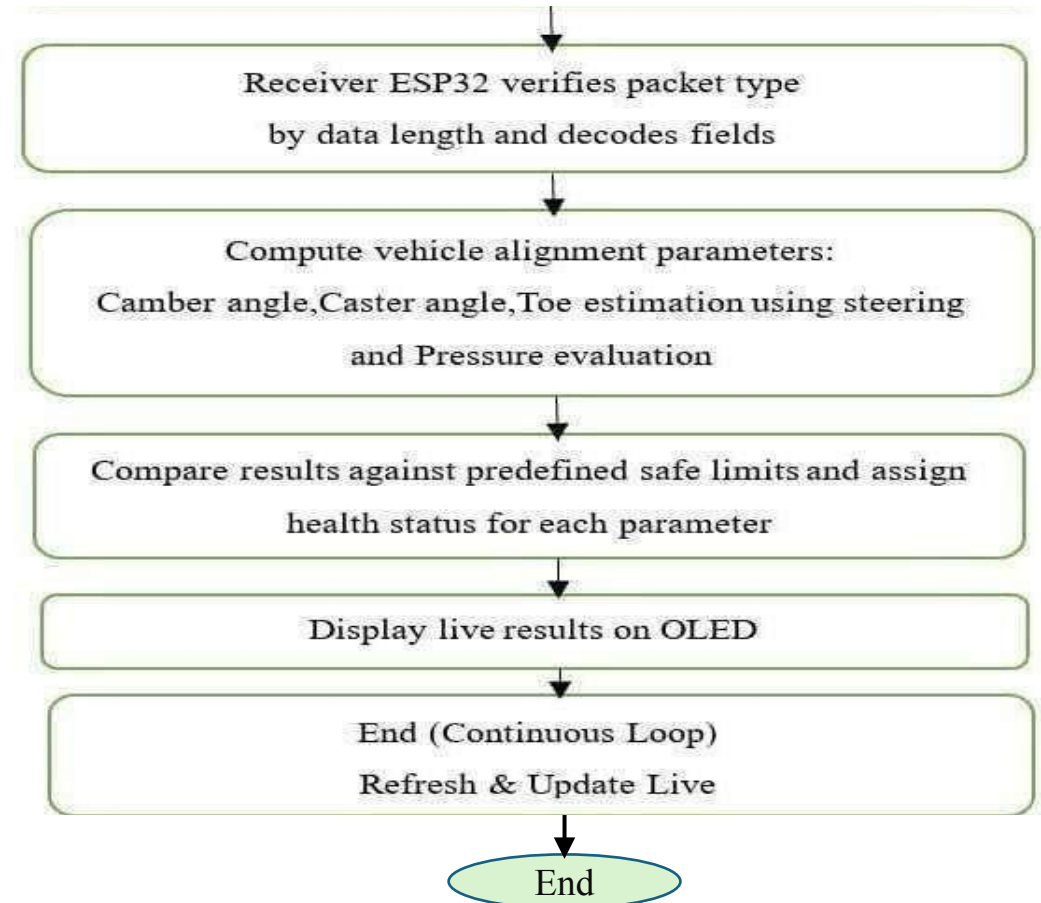
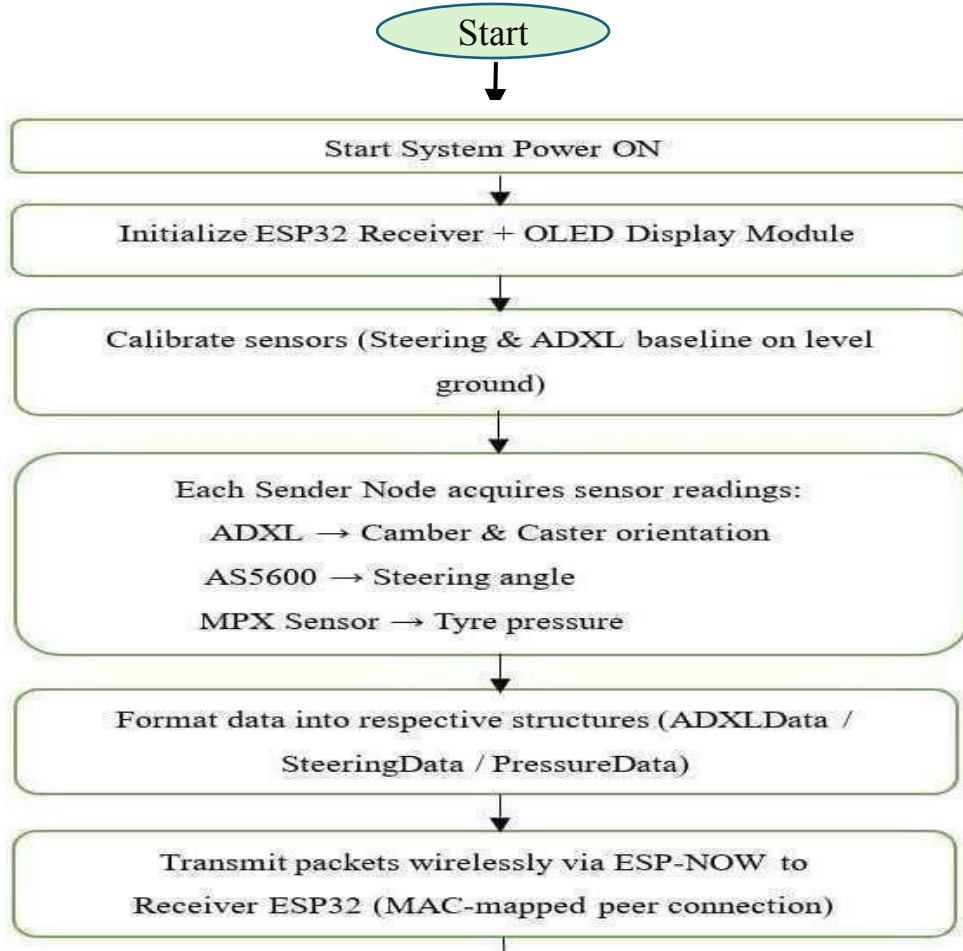
4. Condition Detection

- The system checks for Misalignment (camber/caster/toe faults).
- It also detects Pressure Issues (Low/High) and Final overall status is generated.

5. Result Display

- The OLED shows alignment status, pressure condition, and key sensor readings clearly on one screen.

IMPLEMENTATION



HARDWARE REQUIREMENTS

1. ESP32 Development Board
2. ESP32C3 board
3. ADXL345 Sensor
4. AS5600 Sensor
5. MPX5700AP Sensor
6. OLED display
7. Li-ion Battery
8. TP4056 charging module
9. MT3608 Boost converter

HARDWARE REQUIREMENTS

1. ESP32 development board

- The ESP32 is a dual-core microcontroller with built-in Wi-Fi, Bluetooth, multiple ADCs, and high processing power (240 MHz).
- It is chosen over Arduino UNO because of its faster speed, larger memory, wireless connectivity, and low-power features.
- In this project, it processes sensor data, runs alignment algorithms, and handles display output.

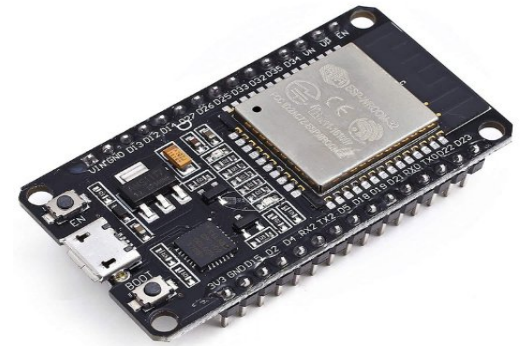


Fig1: ESP32 development Board

2. ESP32C3 Board

- A compact Wi-Fi-enabled microcontroller with very low power consumption.
- Supports ESP-NOW, allowing fast wireless data transfer between modules.
- In this project, it collects sensor data (ADXL345, pressure, steering) and processes the data.
- Sends processed data wirelessly to the receiver for display and final alignment and pressure results.

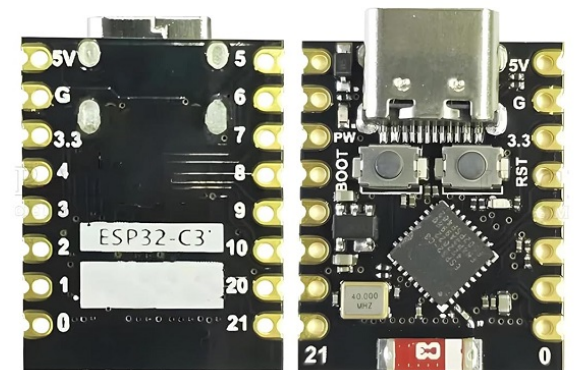


Fig2: ESP32C3 Board

3. ADXL345 Sensor

- ADXL345 is a 3-axis accelerometer that measures tilt, vibration, and orientation. It communicates through I2C, making it easy to interface with microcontrollers.
- In this project, it measures camber and caster angles of the vehicle wheel by sensing tilt along X, Y, Z axes.
- Sends this data to the ESP32C3 for real-time alignment analysis.



Fig3: ADXL345 Board

4. AS5600 Sensor

- The AS5600 is a magnetic rotary position sensor that provides high-resolution steering angle measurement using I²C.
- It detects the angular position of a magnet placed on the steering shaft and converts it into a digital value.
- In this project, the AS5600 is used to measure toe angle and steering reference position, which are crucial for calculating Ackermann geometry and determining whether the wheel alignment is correct or needs adjustment

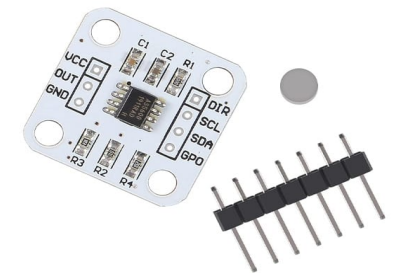


Fig 4:AS5600 Sensor

5. MPX5700AP Pressure Sensor

- The MPX5700AP is an analog pressure sensor designed for automotive use, capable of measuring pressures from 15 to 700 kPa.
- Since vehicle tires typically operate between 200–350 kPa, this sensor is well-suited for tire pressure monitoring
- This makes it a better choice than BMP180, which only supports low atmospheric pressures



Fig 5: MPX5700P

6. OLED Display (128×64, I²C)

- The OLED display offers 128×64 resolution, high contrast, low power consumption, and supports both text and graphics.
- It is selected over LCD due to its sharp visibility, compact size, and no need for backlight.
- In this project, it displays camber, caster, toe, yaw deviation, and alignment status in real time.
- Its fast response makes it ideal for portable, handheld monitoring devices.



Fig 6: OLED display

7. Li-ion Battery(18650/3.7V)

- Rechargeable, lightweight, and high-capacity power source.
- Provides stable voltage suitable for portable embedded systems.
- In this project, It will powers the entire sender and receiver module including ESP32C3, ESP32 and sensors. Ensures the module works portably when attached near the wheel.



Fig 7: Li-ion Battery

8. TP4056 Charging module

- A lithium battery charging module with built-in protection (over-charge, over-discharge).
- Uses a micro-USB input for easy charging.
- In this project, It will charges the Li-ion battery safely and Protects the battery and the entire sender module during charging cycles.



Fig 8: TP4056 module

9. MT3608 Boost convertor

- A DC-DC converter that boosts low voltage (e.g., 3.7V battery) to higher stable voltage (e.g., 5V).
- Has an adjustable output using a small onboard potentiometer.
- In this project, It will Steps up the battery voltage to 5V to power the ESP32C3, ESP32 and other sensors reliably.
- Ensures stable operation even when battery voltage drops during discharge.



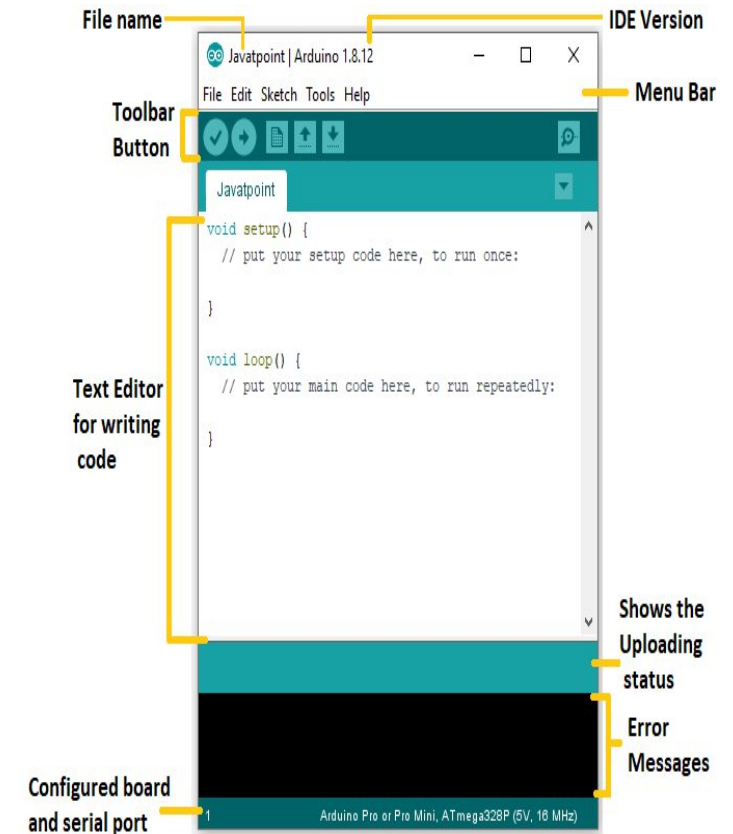
Fig 9: MT3608 Boost convertor

SOFTWARE REQUIREMENTS

Arduino IDE – Platform for writing and uploading the code to the Arduino board.

C/C++ – Programming languages used for sensor integration and data processing.

Sensor Libraries – Libraries for MPU6050, Pressure, OLED, Steering sensor etc.



MATHEMATICAL MODEL

1. Camber Angle

- Camber is the angle between the vertical axis of the wheel and the vertical axis of the vehicle when viewed from the front or rear.

Wheel Camber (tilt of wheel in X-Z plane):

$$\theta_{camber} = \arctan\left(\frac{a_x}{a_z}\right)$$

Where,

a_x = acceleration in X (side tilt)

a_z = acceleration in Z (gravity reference)

Multiply by 180/PI to convert radians → degrees.

2. Castor Angle

- Caster is the angle between the steering axis (line through upper and lower ball joints or struts) and the vertical axis of the vehicle, measured from the side view.

Wheel Caster (tilt in Y-Z plane):

$$\theta_{caster} = \arctan\left(\frac{a_y}{a_z}\right)$$

Where,

a_y = acceleration in Y (longitudinal tilt))

a_z = acceleration in Z (gravity reference)

Multiply by 180/PI to convert radians → degrees.

3. Toe Angle

- Toe is the angle between the direction the wheels point and the longitudinal axis of the vehicle, viewed from above.

$$\theta_{toe} = (angle - refSteering) \times \frac{360}{4096}$$

Where,

AS5600 gives 4096 steps/rev $\rightarrow$ each step = 0.0879° .

Compares current steering angle with calibrated reference.

Detects Toe-In (negative) or Toe-Out (positive).

4. Yaw Rate (Gyro , Ackermann Geometry)

- Yaw rate is the rate at which a vehicle rotates about its vertical axis (z-axis), expressed in degrees per second ($^\circ/s$) or radians per second (rad/s).
- Positive yaw rate $\rightarrow$ vehicle turning left.
- Negative yaw rate $\rightarrow$ vehicle turning right

$$YawRate_{expected} = \frac{(\text{toe in radians})}{L}$$

Where,

L = wheelbase (m)

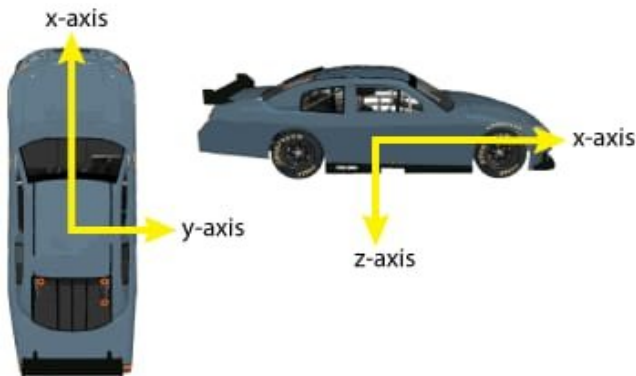
5. Yaw Deviation

- Yaw deviation is the difference between the expected yaw rate (from steering geometry) and the actual yaw rate (measured by gyroscope).

$$\Delta_{yaw} = |\text{YawRate}_{\text{measured}} - \text{YawRate}_{\text{expected}}|$$

Where,

Yaw Threshold = 0.5 → Maximum allowable deviation in °/s

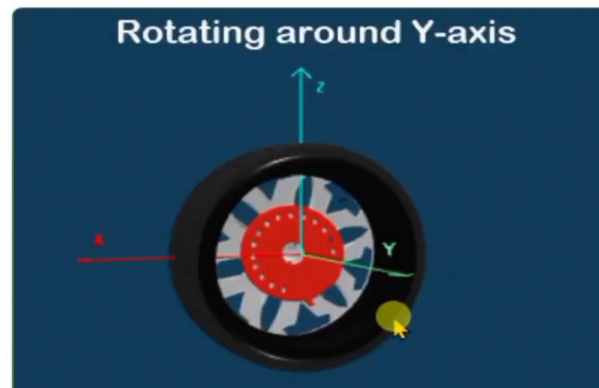


6. Thresholds for Alignment Check

- Camber OK Range: -1° to $+1^\circ$
- Caster OK Range: 2° to 5°
- Toe Threshold: 0.5

--> acceptable deviation from standard toe (0.125°)

- Yaw Threshold: 0.5



RESULTS

Table 1. Normal Wheel Alignment with Normal Tyre Pressure

Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+0.12	+3.10	+0.08	0.5	230	Aligned
2	+0.18	+3.05	-0.05	-0.8	232	Aligned
3	-10	+3.20	+0.12	1.2	228	Aligned

Table 2. Normal Wheel Alignment with Abnormal Tyre Pressure

Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+0.12	+3.1	+0.08	0.5	480	Pressure Issue
2	+0.15	+3.0	+0.05	0.0	495	Pressure Issue
3	+0.10	+2.9	+0.07	-0.4	460	Pressure Issue

RESULTS

Table 5.3 Wheel Misalignment with Normal Tyre Pressure

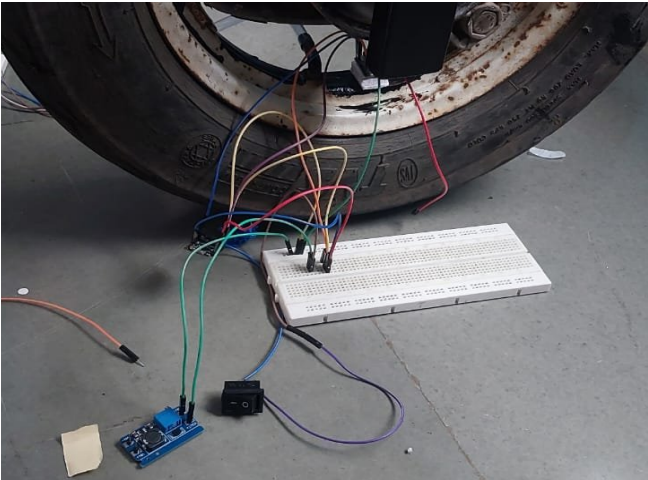
Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+1.8	+4.9	+0.6	0.8	235	Misaligned
2	-1.5	+5.2	-0.7	-0.4	230	Misaligned
3	+2.2	+4.6	+0.9	0.6	240	Misaligned

Table 5.4 Wheel Misalignment with Steering Issue

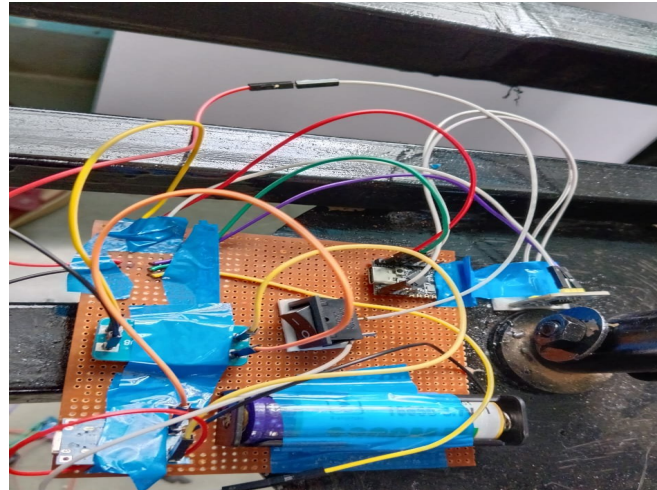
Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+132	-148	+1.6	+68	240	Misaligned
2	-121	+155	-1.8	-72	238	Misaligned
3	+145	-162	+2.1	+74	242	Misaligned

RESULTS

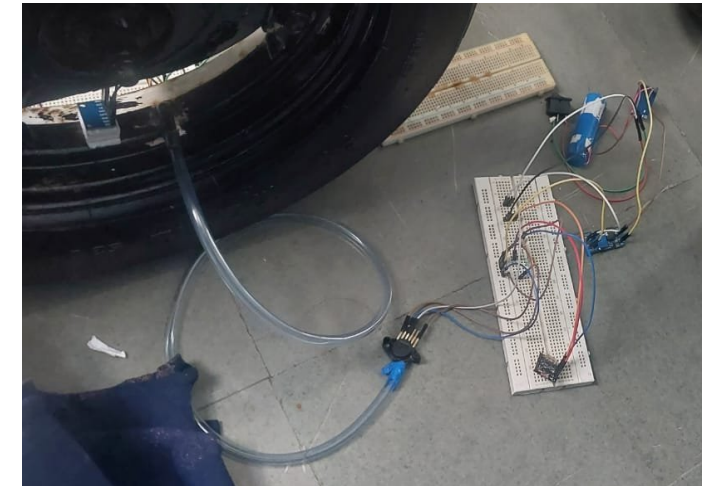
ADXL345 Module



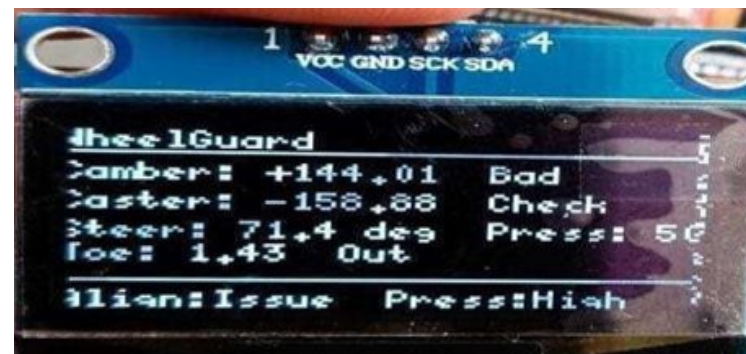
AS5600 Module



MPX5700AP Module



OLED Display



Working Model



CONCLUSION

- The system ensures early detection of wheel misalignment and tire issues enhancing vehicle safety, performance, and efficiency.
- Combines multiple sensor inputs for accurate assessment.
- Offers a practical solution using embedded and IoT technologies.
- Sets the foundation for a low-cost, real-world applicable system.

FUTURE SCOPE

1. Mobile App Integration

- The system can be upgraded to send real-time alignment, camber, caster, toe, and tire-pressure results to a mobile application through Bluetooth/Wi-Fi for easy monitoring.

2. Cloud-Based Data Logging

- Wheel alignment and pressure histories can be stored in the cloud, allowing long-term tracking, maintenance planning, and vehicle-health analysis.

3. AI-Enhanced Prediction & Insights

- Artificial Intelligence can be integrated to analyze past sensor data and predict early signs of misalignment, abnormal pressure trends, or potential tire-wear issues before they become critical.

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